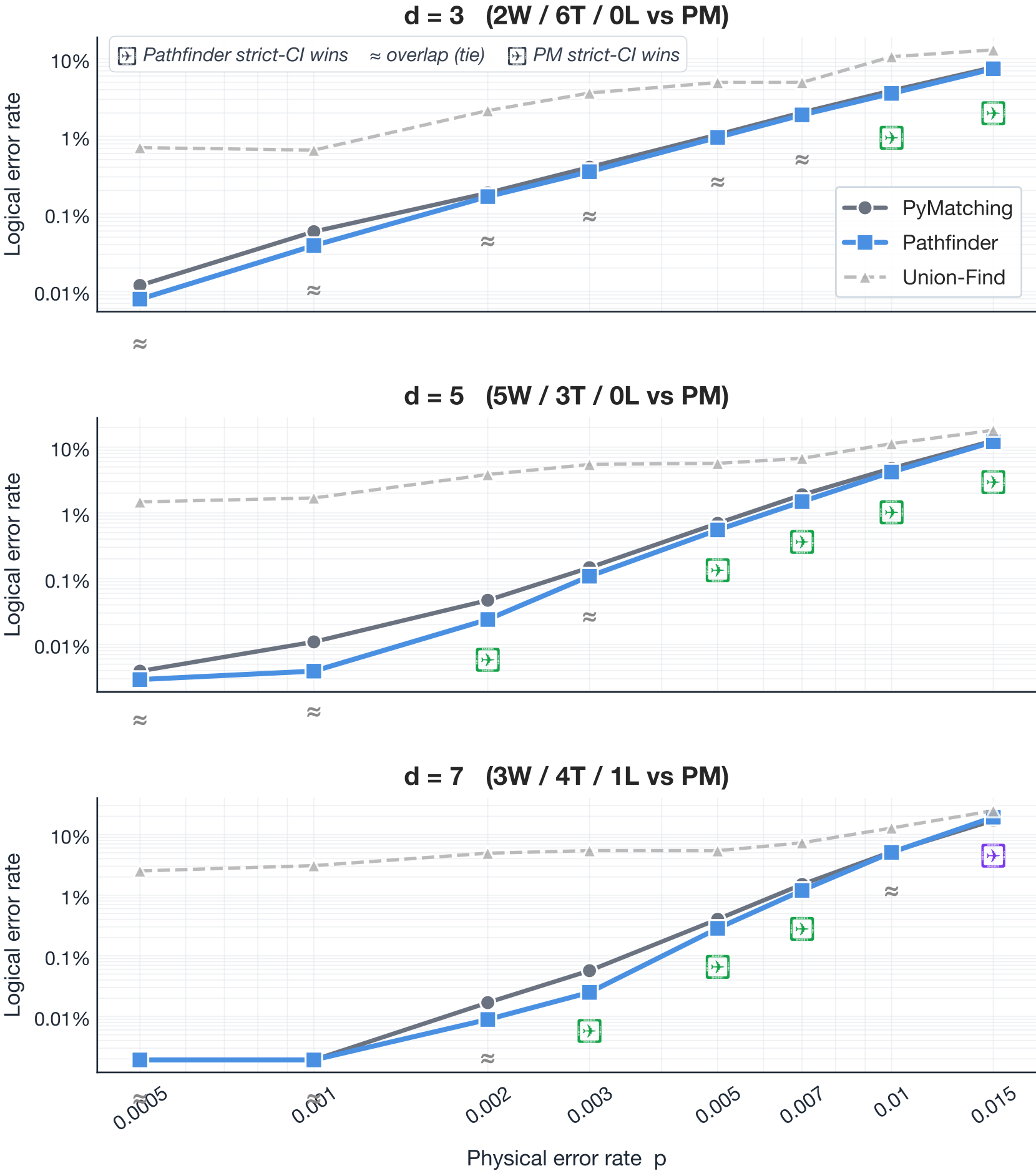


Pathfinder vs PyMatching at  $d \in \{3,5,7\}$ : 10 strict-CI wins / 13 statistical ties / 1 strict-CI loss (3-param noise, 100K shots/point)



Strict-CI test uses 95% Wilson intervals. Pathfinder's only strict-CI loss is at  $d=7, p=0.015$  — the above-threshold regime where the surface code can no longer suppress errors and PM's combinatorial structure is provably near-optimal (same physics as the  $d=9, p=0.015$  result of Fig 11).